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Contact information
Pieter.Kempeneers (at) ec.europa.eu

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Preface

In times marked by global crises, Europe's resilience–digital, societal, and environmental–has never been more vital. Being resilient means safeguarding our societies and our future in the digital space race: upholding our laws, values, and sovereignty in how data are accessed, shared, and protected. With the adoption of the European Union's Data Act (September 2025) and the world's first comprehensive AI Act, Europe has taken bold steps to ensure that innovation and competitiveness go hand in hand with trust, fairness, and fundamental rights. Together, these acts set the stage for a distinctly European way of governing the digital future, ensuring that Big Data and AI serve not just technology, but people, society, and the planet.

Big Data from Space (BiDS) is not just about observing the Earth. It is about enabling smarter, faster, and fairer decisions to meet the challenges of our time. From climate change and civil security to sustainable competitiveness and digital innovation, the insights we derive from spaceborne and terrestrial data are essential for evidence-informed action. In this sense, BiDS'25 was not only about advances in science and technology, but also about ensuring that data-driven innovation supports Europe's ability to address societal and security challenges.

Building on the success of the initiatives introduced in the last edition, BiDS'25 organised 23 satellite events, 28 demos, lightning talks, and birds-of-a-feather sessions. As in BiDS'23, code sprints were held with OSGeo and Pangeo to promote collaborative open-source software development. BiDS'25 also introduced new formats and perspectives, including an award session for the best start-up ideas. It demonstrated how scientific breakthroughs can evolve into entrepreneurial solutions and strengthen Europe's competitiveness.

Another novelty for this year's conference was the panel discussions that aligned with our mission of connecting deep tech with deep purpose. On Wednesday, October 1, "Society at Risk: Challenges and Priorities in the Space-Data Age", explored critical questions at the intersection of technology and societal well-being. On Friday, October 3, the final day of the conference, "From Raw Data to Real Decisions: Systems that Work in the Space-Data Age" focused on bridging raw data to actionable decisions through resilient infrastructure, AI, cybersecurity, and ethical frameworks–setting priorities for future research, policy, and societal impact. These conversations reflected our commitment to turning planetary-scale data into scalable solutions for global challenges.

Latvia's space sector, a partner of ESA with a rich historical heritage and modern technological strengths, brought unique value to this conference. With the support of the Investment and Development Agency of Latvia (LIAA), a B2B networking event and industrial exhibition highlighted Latvia's growing role in the European space economy.

The figures of BiDS'25 demonstrate both its reach and dynamism: two weeks before the start, 621 participants had registered. With more than 40 participants on average, the satellite events were fully booked, and 152 developers had registered for the code sprints. A total of 132 submissions were received from 34 different countries, including papers, demos, satellite, and award events. Each submission was reviewed by at least two experts from the Programme Committee. Of the 75 papers submitted, 42 were accepted as oral presentations and 21 as posters. While Earth Observation played a central role in this edition, BiDS covers the full spectrum of space domains–navigation, science and communications–and this breadth remains essential to its identity. The presentations were organised across seven thematic sessions: Towards Digital Twins: Integrating Data, Models, and Insight; Data Cubes: Advances and Applications; FAIR Workflows; Data Infrastructures & Services at Scale; GeoAI & Geospatial Intelligence; Optimising Processing from Edge to Cloud; Societal Applications: Risk, Resilience, and Resource Monitoring.

This year's BiDS'25 keynote talks captured the breadth and urgency of the challenges at the intersection of data, science, and society. Thomas Brunschwiler (IBM Research Europe, Switzerland) explored how tokens and embeddings are emerging as a new lingua franca for AI-driven Earth Observation, reshaping environmental science and data workflows through foundation models. Lynn Dudenhöfer (Senior Intelligence Professional) highlighted how the fusion of open-source intelligence and EO data is becoming indispensable for addressing rapidly evolving threats, from transnational organised crime to geopolitical instability, in a security landscape increasingly shaped by GenAI. Rosa M. Badia (Barcelona Supercomputing Center, Spain) connected the dots from Europe's world-class supercomputing infrastructure to domain-specific and FAIR application workflows and digital twins. She showed how HPC, AI, and data analytics can together deliver actionable solutions for society, with a focus on geohazards and risk mitigation. Together, these keynotes set the tone for BiDS'25: advancing science and technology while ensuring that innovation is directed towards

resilience, security, and sustainability.

Altogether, BiDS'25 offered a vibrant mix of science, technology, policy, and entrepreneurship. It stood at the intersection of climate resilience, societal security, and digital sovereignty, reminding us that responsible innovation is no longer optional. By bringing together diverse communities, BiDS'25 reaffirmed its role as the place where Europe and the world set the course for data-driven resilience and sustainability. Several contributions pointed to applications with direct impact on resilience and decision-making globally, as well as for EU Member States and institutions, showing a strong bridge between research and operations. This resonated with the European strategic context, where flagship programmes such as the EU Space Programme–covering Copernicus, Galileo, and other space infrastructures–together with key policy frameworks like the EU Strategic Compass and the EU Space Strategy for Security and Defence, supported the Union's ambition for leadership and strategic autonomy.

We express our sincere gratitude to the Programme Committee members for ensuring the excellence of this programme and these proceedings. Our local partner, the Ministry of Education and Science of Latvia, played a pivotal role in hosting BiDS'25. Thanks to their support, we gathered at the inspiring National Library of Latvia, and the University of Latvia hosted the satellite events. BiDS'25 again showcased the strong engagement of its three organising entities–SatCen, ESA, and JRC–which remain at the forefront of innovation and central to the themes of this edition. Ultimately, the outcomes of BiDS'25 extend beyond the scientific community to operational actors–including those working in security and decision-making–ensuring that innovation reaches end-users and has tangible impact.

Pieter Kempeneers, Stefanie Lumnitz,
Sergio Albani

GEO-OPEN-HACK: AN INITIATIVE ON BIG GEOSPATIAL DATA PROCESSING WITH OPEN COMPUTING INFRASTRUCTURE AND OPEN TOOLS

Milutin Milenković[1], *Giuseppe Amatulli*[2], *Tushar Sethi*[2], *Raymond Oonk*[3], *Anne Fouilloux*[4], *Tina Odaka*[5], Michele Claus[6], Valentina Premier[6], Antonio Fonseca[2], Pieter Kempeneers[7], Francesco P. Lovergine[8], Leandro Parente[9], Yu-Feng Ho[9], Johannes Heisig[10], Dainius Masiliunas[11], Wolfgang Wagner[12], Edzer Pebesma[10], Patrick Griffiths[13], Jan Verbesselt[14], Tom Hengl[9], Ian McCallum[1], Steffen Fritz[1]

[1]IIASA,Austria, [2]Yale University USA, [3]SURF, The Netherlands [4]Simula, Norway, [5]Ifremer, France, [6]Eurac, Italy, [7]JRC, Italy, [8]CNR IREA, Italy, [9]OpenGeoHub, The Netherlands, [10]Uni Münster, Germany [11]Wageningen Uni, The Netherlands, [12]TU Vienna & EODC, Austria, [13]ESA, Italy, [14]BELSPO, Belgium

ABSTRACT

To maximize the potential of big geospatial data streams, they shall be processed on high-performance computing (HPC) infrastructure, whereas local code shall be adopted to use this infrastructure efficiently. This paper presents a non-profit initiative, GEO-OPEN-HACK, that aims at lowering the technical barrier for environmental scientists through one-week hackathons on open-source HPC processing approaches, such as openEO, Pangeo, and OEMC. We present the above solutions, hackathon setup, and analysis based on the pre-event questionnaire about the participants' big data projects.

Index Terms— Geocomputation, Education, HTC

1. INTRODUCTION

A full exploration of big geospatial data, originating mainly from open satellite image archives such as those of the European Commission's Copernicus and the NASA-USGS Landsat programmes, requires both high storage and high computing capacity that no longer fit in a local workstation used for Earth Observation (EO) studies. Thus, studies typically rely on high-performance computing (HPC), which enables parallel processing across multiple processors and nodes, to dramatically accelerate their data processing [6, 5].

HPC also comes with technical complexity in its storage and computing systems. To address this, several higher-level open-source engines for geospatial workflows have been introduced recently. For example, openEO masks out code parallelization, unifies the interaction with different backends, and also supports processing user-defined functions [7]. Pangeo provides server-side computation via lazy evaluation, relying on Python tools (xarray) to handle multidimensional datasets and Dask for parallelization [1]. The Open Earth Monitor Cyberinfrastructure (OEMC) project is another open initiative that supports existing solutions with additional open tools and data.

Several hackathons, workshops, and summer schools feature Geospatial data processing with open tools. Spatial Ecology is organizing an annual event on geocomputation and machine learning (ML) for environmental applications. OpenGeoHub Foundation is also organizing yearly summer schools on data science for EO. Those events are either on the introductory or intermediate level, with a primary focus on building open geospatial workflows. At the last BiDS2023 conference, an openEO and Pangeo tutorial was given, but rather as a half-day sprint. Thus, there is a need for an event where intermediate- and advanced-level participants can learn about different HPC approaches and look at them through the prism of their use cases.

This paper presents the GEO-OPEN-HACK, a non-profit initiative focused on transferring knowledge about the advanced processing of big spatial data using open tools in an HPC environment. We outline the objectives and setup of GEO-OPEN-HACK-2024, and provide an overview of HPC processing with openEO, Pangeo, and OEMC.

2. GEO-OPEN-HACK OBJECTIVES AND SETUP

GEO-OPEN-HACK-2024 was a give-and-take, non-profit initiative introduced under the OEMC project to: (a) lower the barrier and transfer knowledge to users dealing with big geospatial data analytics, i.e., the give part, and (b) foster the co-creation of open OEMC geospatial data and tools through user feedback, i.e., the take part. The event took place from June 24 to 28, 2024, at the International Institute of Applied Systems Analysis (IIASA) in Laxenburg, Austria. The focus was on presenting HPC approaches with different levels of technical complexity so that the participants were enabled to: (a) judge whether their big data problem is suitable for

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HPC, (b) determine what would be the most appropriate HPC approach for their big data problem, and (c) process their data on HPC. This was an advanced-level hackathon, ideal for early-career researchers, scientists, and professionals who already had intermediate skills in Linux, bash, Python, and R. A basic familiarity with geoinformation processing and EO was also required. This was a one-week event featuring a keynote speech, a tutorial by teachers, a big-data story by a teacher, and a hands-on session led by participants on their own data. Here is the detailed agenda and links material.

The participants worked on three backends. They mostly worked on the Spider cluster, which is a part of the Dutch national research infrastructure, SURF. The cluster comprises 42 nodes, providing over 3,000 cores with 8 GB RAM and 80 GB local SSD scratch disk space per core. Furthermore, Spider offers PB-scale internal storage with CephFS filesystem and network uplink of 1200 Gbit/s. With openEO and Pangeo, participants also worked with the Copernicus Data Space Ecosystem (CDSE) and the European Open Science Cloud (EOSC) backends.

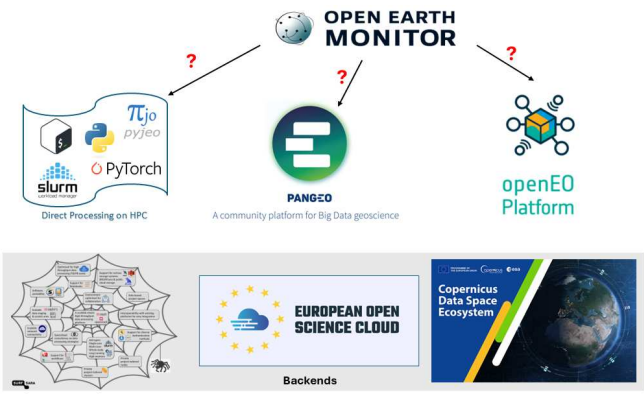

Fig. 1. The GEO-OPEN-HACK-2024 setup with three different HPC processing approaches and three backbends.

3. GEODATA PROCESSING APPROACHES

3.1. Direct Processing on HPC

During the direct processing on HPC, the participants had to set up an HPC environment for efficient geo-data processing. This involves organizing a logical folder structure, preparing outputs, and utilizing submission scripts, such as sbatch, to efficiently manage tasks . Proper directory and alias setup allows users to monitor jobs, handle errors, and reprocess failed tasks easily. Bash scripting automates these processes, ensuring repeatability and minimising human error. Raster processing, resampling, modelling, and landscape analysis become feasible even for continental-scale studies. Techniques such as splitting datasets, parallel resampling, and running GIS commands in batch are essential. Ultimately, such an HPC setup transforms geospatial analysis from a bottleneck into a streamlined, robust operation.

3.1.1. GDAL, pktools and GRASS GIS on HPC

In the hands-on session, multiple scripting techniques were introduced using GDAL, pktools, and GRASS. These tools provided fast, flexible, and scalable features and functions for raster-based analysis with a Bash command environment. Moreover, parallelization is accomplished by dividing a large geospatial dataset into regular tiles, conducting the operation on each tile with a dedicated CPU, and finally mosaicing them back to the original extent. A concise overview and description of the script used in the hackathon for the above steps are given on the BASH tutorial and the GRASS tutorial pages.

3.1.2. pyjeo and HPC

pyjeo is an open source library for geospatial image processing in Python [4]. It has been implemented in the Joint Research Centre of the European Commission and is distributed under the GPLv3 license. The design of pyjeo is based on a C/C++ implementation with Python bindings that are made available through the Simplified Wrapper and Interface Generator (SWIG [2]). Parallel processing is supported via multithreading using the OpenMP API in C/C++. Additionally, high-throughput computing can be achieved through a tiling mechanism when combined with a job scheduler. Unlike other libraries such as Dask, the tile-based orchestration, including the merging step, is not automated and is not part of the pyjeo library. It also relies on a job scheduler such as HTCondor to launch the individual jobs that process the different tiles. In [3], the authors show different strategies using pyjeo for the parallel processing of geospatial data in a cloud computing infrastructure.

3.2. Pangeo Approach

The Pangeo ecosystem provides an open-source, cloud-, and HPC-ready platform for scalable geoscientific and climate data analysis. Built on modern Python tools such as xarray, Dask, and Zarr, Pangeo facilitates the analysis of multi-dimensional datasets that exceed in-memory capacity. Within the GEO-OPEN-HACK event, Pangeo was introduced as a framework that supports interactive, scalable workflows, embodying FAIR (Findable, Accessible, Interoperable, Reusable) and reproducible science principles.

Pangeo involves lazy evaluation and parallel execution using Dask. This allows users to work efficiently with large volumes of data stored in cloud-optimized formats (COG and Zarr). Participants engaged in hands-on workflows implemented in Jupyter notebooks deployed on Dask clusters, exploring essential EO data processing tasks including subsetting, aggregation, statistical analysis, and visualization.

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Two Pangeo deployment setups were demonstrated: one on EOSC and another on the Spider cluster. The latter leveraged Apptainer containers and the hpc-container-wrapper framework to execute Pangeo environments efficiently on SLURM. Developed in collaboration with SURF and the Netherlands eScience Center, the Spider deployment built upon open scripts and documentation contributed by Francesco Nattino, Meiert W. Grootes, and Ou Ku. Participants were instructed on launching JupyterLab with Dask via SLURM job scripts and accessing remote sessions using SSH tunneling. For users lacking container access, alternative solutions using micromamba, jupyter-forward, and dask-jobqueue were also demonstrated.

Both deployments emphasized the paradigm shift from downloading data locally to executing code where the data resides—leveraging public S3 buckets and open SpatioTemporal Asset Catalogs (STAC) for direct data access. This approach was particularly beneficial for participants constrained by limited local computational resources. The event further highlighted containerized reproducibility through the use of Docker and Binder, enabling the creation and sharing of fully executable environments.

3.3. openEO Approach

openEO is an open-source framework that standardizes and streamlines the access, processing, and analysis of EO data across heterogeneous backends. Traditional EO workflows often involve time-consuming steps, including data discovery, download, and platform-specific pre-processing. This poses challenges in scalability and interoperability, especially when integrating multiple data sources. openEO addresses these limitations by defining a unified API that abstracts the underlying data infrastructures and exposes consistent functionality through client libraries in various languages, including Python, R, and JavaScript.

During the hackathon, we showcased the capabilities and usability of the openEO framework. Our demonstration used the openEO Python client within Jupyter notebooks to connect to the CDSE backend. Participants were guided through essential steps, including library import, authentication, and backend connection. Limited memory on the CDSE Hub caused occasional kernel crashes, which were mitigated by reducing data size. Additionally, we introduced the concept of datacubes, a central feature of openEO that structures EO data in a spatiotemporal grid. This model enables server-side declarative processing, significantly streamlining complex analyses and enhancing reproducibility and cross-platform integration in EO applications.

3.4. OEMC Approach

OEMC software and data products were also presented at the hackathon to gather user feedback and foster co-development.

3.4.1. GlobalPointTools

GlobalEarthPoint is an open-source software library for accessing large geospatial point datasets stored in cloud environments such as S3. Python and R bindings enable efficient (interactive) queries and downloads of data stored in cloud-optimized formats (e.g., partitioned GeoParquet). The software supports lazy evaluation for high-performance data retrieval through technologies like Apache Arrow and Polars.

3.4.2. OEMC ML Deployment and Central app

Besides organizing and enhancing the accessibility of existing EO and in-situ data, the OEMC project is producing European-wide [5] and global datasets [6] at a 30-m spatial resolution, tailored to meet the needs of the use cases. The production of these datasets is running via an ML pipeline deployed in an in-house HPC/HTC infrastructure hosted by OpenGeoHub Foundation. With more than 1,000 CPU threads across 14 high-density nodes and 3 Petabytes of storage space, the infrastructure runs a suite of open source solutions, including `SeaweedFS` as object storage, `SLURM` as workload manager, and `Docker` as containers manager. The ML algorithms/methods rely on `scikit-learn` and utilize spatiotemporal overlay and predictions via `scikit-map`. Feature selection, hyperparameter tuning and the training of the final ML models run in a single node (96 CPU threads and 1 TB of RAM), while the prediction/inference of billions of 30-m pixels run in `SLURM`, where each chunk of input data (ex. 4000x4000 pixels) is assigned to an processing node by an embarrassing parallel strategy. Predictions of tree-based models were sped up by compiled models based on `tl2cgen` and `lleaves` libraries. The OEMC ML Deployment proved to be reliable in a production environment once all layers, produced in the context of the project, are publicly available as open data (CC-BY) at OEMC Central App, OpenLandMap STAC, EcoDataCube STAC, and Zenodo.

4. RESULTS

4.1. Hackathon Participants

The hackathon had 20 on-site and 21 online participants from 17 countries, with the majority coming from Europe. Most of them were PhD students ($\tilde{4}6\%$), followed by early-career researchers ($\tilde{2}5\%$), and postdoctoral researchers ($\tilde{1}8\%$); the rest comprised master's students, professors, and employees from the private sector. Most of them consider themselves ecologists or environmentalists, but there were also remote sensing scientists, geographers, hydrologists, and economists (Fig. 2).

The questionnaire also revealed that every fourth participant had prior experience with HPC processing. Nevertheless, most of them have also been using open-source tools and

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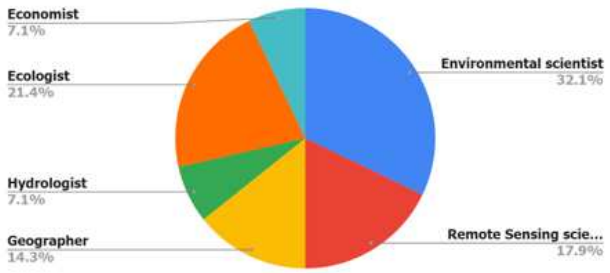

Fig. 2. The expertise of the hackathon participants.

Python modules, such as GDAL, Rasterio, NumPy, Geopandas, QGIS, GRASS, PyTorch, PKTools, Pandas, and Scikit-learn.

4.2. Perception of Big Data

Almost half of the participants ($\tilde{4}6\%$) work with geospatial datasets that are stored both online and locally, whereas the rest of the participants work only with data stored online ($\tilde{2}8\%$) or only with data stored locally ($\tilde{2}6\%$). Their locally stored data are primarily in the 10-100 GB range, whereas only about 15% work with locally stored data on a TB scale (Fig. 3). Furthermore, the data were mainly related to the following topics: land monitoring ($\tilde{2}0\%$), hydrology ($\tilde{1}6\%$), agriculture ($\tilde{1}4\%$), vegetation ($\tilde{1}1\%$), although topics such as hazards, economics, climate, soil, urban, health, and biodiversity were also covered.

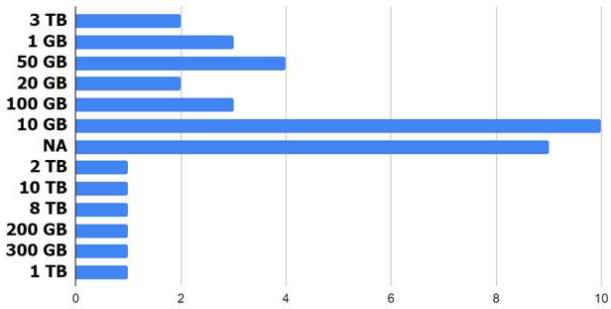

Fig. 3. What is the size of your local geospatial dataset that you work with?

5. DISCUSSION AND CONCLUSIONS

Despite the capabilities of the HPC approaches, participants encountered challenges related to environment configuration, Dask performance tuning, and conceptual adaptation from serial to parallel processing. These insights underscore the importance of streamlined deployment practices and enhanced training resources in lowering entry barriers and fostering the broader adoption of scalable open science workflows.

The pre-hackathon questionnaire, completed by participants, revealed several interesting insights. The participant structure showed that the hackathon primarily attracted researchers and PhD students, with a notably small number of participants from the private sector. However, the geospatial industry and academia should collaborate more, as open-source projects can benefit more from such partnerships. Thus, the next hackathons should focus more on engaging with industry.

The local data volumes typically processed by the participants were 10 TB or smaller. Such data can be readily handled with HPC, suggesting that the barrier is rather in adjusting code for efficient HPC processing. Therefore, presenting and interacting with different HPC processing approaches and backends is fully justified and shall remain the focus for future hackathons.

Finally, it is worth noting that, in addition to a one-time budget allocated by IIASA and the OEMC project for organizing GEO-OPEN-HACK-2024, the event received significant support from the community, including teachers and organizations such as SURF, Spatial Ecology, and OGH. Nevertheless, it is essential to secure future funding sources to ensure its long-term sustainability.

REFERENCES

[1] Abernathey, R. P., Augspurger, T., and et al. (2021). Cloud-native repositories for big scientific data. *Computing in Science Engineering*, 23(2):26–35.

[2] Beazley, D. (2003). Automated scientific software scripting with swig. *Future Generation Computer Systems*, 19(5):599–609.

[3] Kempeneers, P., Kliment, T., and et al. (2022). Parallel processing strategies for geospatial data in a cloud computing infrastructure. *Remote Sensing*, 14(2).

[4] Kempeneers, P., Pesek, O., and et al. (2019). pyjeo: A python package for the analysis of geospatial data. *ISPRS International Journal of Geo-Information*, 8(10). 10.3390/ijgi8100461.

[5] Minarik, R., Hengl, T., and et al. (2024). Soil type (world reference base) map of europe based on ensemble machine learning and multiscale eo data. *researchsquare*. Submited to PeerJ. 10.21203/rs.3.rs-5244083/v1.

[6] Parente, L., Sloat, L., and et al. (2024). Annual 30-m maps of global grassland class and extent (2000–2022) based on spatiotemporal machine learning. *Scientific data*, 11(1):1–22.

[7] Schramm, M., Pebesma, E., and et al. (2021). The openeo api–harmonising the use of earth observation cloud services using virtual data cube functionalities. *Remote Sensing*, 13(6).

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